Prevention of urinary stones with hydration: a randomised clinical trial of an adherence intervention

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Summary

Background Increased fluid intake is universally recommended to decrease the risk of recurrent urinary stones; however, adherence is challenging. The effectiveness of interventions to maintain high fluid intake has not been well studied. We sought to determine whether a multicomponent behavioural intervention programme to promote high fluid intake reduces symptomatic stone recurrence, compared with a control.

Methods In this randomised clinical trial, participants aged 12 years and older with a history of urinary stone disease and low 24 h urine volumes based on current guidelines were enrolled at six academic medical centres in the USA. Participants were randomly assigned in a 1:1 ratio to a multicomponent behavioural intervention designed to promote increased fluid intake or to the control group receiving guideline-concordant care. The intervention consisted of a fluid prescription, financial incentives to adhere to fluid prescription, health coaching to overcome barriers to consuming more fluids, and patient-selected approaches such as text messaging to maintain increased fluid intake. Randomisation assignment was computer-generated remotely, and investigators, treating physicians, outcome assessors, and adjudicators were masked to group assignment. The primary outcome was symptomatic stone recurrence defined as stone passage or procedural intervention for stone(s) during a 2-year follow-up period, analysed in the intention-to-treat population. Secondary outcomes included change in 24 h urine volume, urinary symptoms, radiographic stone recurrence or growth, and a composite outcome of symptomatic stone recurrence, new stone formation, and growth of existing stone(s); hyponatremia requiring hospitalisation was the safety endpoint. This trial is registered with ClinicalTrials.gov, NCT03244189.

Findings Between Oct 26, 2017, and Feb 18, 2022, 1658 participants were randomly assigned to intervention (n=826) and control (n=832) groups (median age 44 years [IQR 29–59]; 946 [57%] female). At a median follow-up of 738 days (IQR 711–778), symptomatic stone events occurred in 154 (19%) participants in the intervention group and 165 (20%) in the control group (hazard ratio 0·96, 95% CI 0·77–1·20). Among these 1658 participants, 1104 (66·6%) were recurrent stone formers. 24 h urine volume increased from baseline in both groups and was higher in the intervention group at months 6, 12, 18, and 24 compared with the control group. Urinary storage symptoms of frequency, urgency, and nocturia were greater in the intervention group versus control at months 6 and 12 but not at other timepoints. There was no difference in stone growth of at least 2 mm or new stones between groups from baseline to end-of-study imaging, and the composite outcome of symptomatic stone recurrence, new stone formation, or stone growth of at least 2 mm was also not statistically different between groups. No episodes of hyponatremia requiring hospitalisation (safety endpoint) were reported; 12 (1%) participants in the intervention group had asymptomatic hyponatraemia versus two (<1%) participants in the control group.

Interpretation A behavioural intervention programme to promote fluid intake for secondary stone prevention did not reduce recurrent stone events but modestly increased urine volume compared with guideline-based care during a 2-year follow-up period.

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See **Comment** page 1123

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Introduction

Urinary stone disease is a common disorder of mineral metabolism marked by episodic painful events[1] that negatively affect physical, social, and emotional health.[2] Guideline-based prevention strategies emphasise high fluid intake as a cornerstone of reducing recurrence risk.[3–5] A Cochrane review[6] included a single randomised controlled trial that showed increased water intake reduced urinary stone recurrence, rated as low-certainty evidence. This seminal trial by Borghi and colleagues,[7] comparing high fluid intake versus no additional fluid intake in 199 participants, demonstrated benefit for increased hydration. Thus, achieving and maintaining high fluid intake could provide safe and effective secondary prevention of symptomatic urinary stones. However, maintaining high fluid intake is a formidable

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See Online for appendix

Research in context

Evidence before this study

We performed a systematic search of PubMed for randomised trials testing interventions to promote adherence to fluid intake for secondary prevention of urinary stone disease. We searched PubMed (from Jan 1, 1995, to June 2, 2025); the detailed search strategy is available in the appendix (p 4). We also searched ClinicalTrials.gov (from inception to June 2, 2025) to identify similar trials in progress or without published results.
We identified 222 publications and five ClinicalTrials.gov listings. Of these, we identified six trials (including one published only as a conference abstract) testing adherence interventions for fluid intake for secondary stone prevention. Of these six trials, two were initiated before 2017, when our Prevention of Urinary Stones with Hydration (PUSH) trial began. The Hidrate Me study (NCT02938884) randomly assigned participants with a history of urinary stone disease and low urine volume to a smart water bottle (Hidrate Spark) versus standard water bottle, facilitating self-monitoring of behaviour in the intervention group.
The results of this trial were reported in 2022. Among 85 participants enrolled, 51 (60%) completed 24 h urine collections at 6 weeks, with a greater mean increase in urine volume over baseline in the smart water bottle group than in the control group (+1·37 [SD 0·94] L per day *vs* +0·79 [SD 0·97] L per day, p=0·04). Stone recurrence was not ascertained as an outcome. The second study, initiated before 2017, was an observational cohort study (NCT01928108) enrolling adults with a history of urinary stone disease. This study compared use of two different smartphone applications to manually track water consumption, also using a behaviour change technique of self-monitoring. Neither the literature search nor ClinicalTrials.gov reported study results. The other four trials of adherence interventions for fluid intake in prevention of urinary stone disease were initiated after PUSH started. The protocol describing the sipIT2 trial (with a primary endpoint of 24 h urine volume) was published in 2024. Two randomised controlled trials identified only via ClinicalTrials.gov compared a smartphone application delivering prevention education versus a control group receiving no stone prevention information.
The sixth trial (comparing the mobile app care plan versus the standard kidney stone follow-up pathway) was identified only via a conference abstract; no ClinicalTrials.gov registration was identified. Current guidance for water intake for secondary stone prevention is based on a Cochrane Systematic Review (2020) that found a single randomised controlled trail (Borghi, 1996) comparing the effects of high water intake versus low water intake for secondary urinary stone disease prevention. Cochrane assessed this randomised controlled trial as low-certainty evidence.

Added value of this study

The PUSH trial advances the field by testing an adaptive, multicomponent behavioural health approach to promoting fluid intake adherence in a large population of patients with urinary stone disease and low urine volume. The PUSH intervention is grounded in contingency management, leveraging a loss-framed financial incentive, along with an adaptive structured problem-solving intervention that is responsive to participant non-adherence. In addition to examining 24 h urine output, PUSH is the first adherence study to assess the clinical endpoint of urinary stone recurrence.

Implications of all the available evidence

Very little evidence is available to guide clinicians about the best strategies that will increase and sustain high fluid intake for secondary urinary stone disease prevention. The PUSH trial results suggest that for many patients with urinary stone disease and low urine volume, it might be difficult to sustain high fluid intake and thereby reduce stone recurrence. Taken together, these results suggest that investigators might need to focus on alternative adherence strategies and secondary prevention strategies that go beyond simply increasing fluid intake.

challenge: the average increase in urine volume after counselling by physicians is only 300 mL per day.[8] Therefore, the Prevention of Urinary Stones with Hydration (PUSH) study focused on improving adherence to high fluid intake, testing an intervention grounded in behaviour change theory, and focusing on addressing common barriers to high fluid intake.

Barriers to achieving and maintaining high fluid intake are manifold: patients frequently identify lack of awareness of volume consumed or intake goals, forgetting to drink, and perceived need to drink as major impediments.[9] Other barriers include practical concerns such as lack of access to water, lack of bathroom access, or competing time demands. Similar types of barriers to behaviour change exist for many diet-related health conditions, including obesity, for which the United States Preventive Services Task Force recommends multi-component behavioural interventions to improve weight status and reduce the incidence of type 2 diabetes.[10] The PUSH study intervention is designed as an adaptive, multicomponent intervention leveraging multiple behaviour change techniques[11] to promote adherence to fluid intake. As part of a contingency management approach, the PUSH study leverages financial rewards for behaviour change, which is a well validated and widely accepted way to incentivise a broad range of health behaviours, including reducing substance use, increasing physical activity, and promoting weight loss.[12] Although contingency management is effective at eliciting behaviour change, it might be more effective when utilised as part of a multicomponent behaviour change strategy.[13] In what follows, we describe the key components of the PUSH intervention with the use of the standard taxonomy of behaviour change techniques (BCT)[11] and the Behavior Change Intervention Ontology number (BCIO, found at bciosearch.org). In the PUSH

For more on BCIO see https://www.bciontology.org/search

trial, the primary intervention to support increased water intake was providing a smart water bottle and loss-framed financial incentives (providing aversive material consequence for BCT; BCIO:007243). Additional intervention in the form of health coaching (structured problem-solving—guide how to perform behaviour BCT; BCIO:007050) was available to individuals who consistently did not meet water intake targets over any 2 week period in the study, to help them to overcome practical barriers to fluid intake. This type of adaptive intervention strategy is common in health behaviour change, with additional support provided if a patient does not meet goals.[14] Health coaching and other low-touch approaches are commonly added to BCIs when people are not meeting their goals.[15] The rationale for adding structured problem-solving for participants falling short of their goals was intended to provide assistance for identifying and solving barriers to meeting the prescribed water intake.

The PUSH study selected recurrent symptomatic urinary stones as the primary endpoint, rather than the more proximate outcome of increased fluid intake (or 24 h urine output, which is the guideline-based clinical surrogate). The efficacy of behavioural interventions should ideally be determined by clinically important outcomes that are meaningful to patients, because surrogate outcomes might overestimate the benefits of the intervention.[16] PUSH tested the hypothesis that this adaptive, multi-component behavioural intervention to promote fluid intake would be more efficacious to increase urine volume and reduce recurrent symptomatic stone events than a control group receiving guideline-based care from their usual physicians.

Methods

Study design

The PUSH trial was an investigator-initiated, randomised controlled trial conducted between Oct 4, 2017 and April 16, 2024, at six medical centres in six US states that participate in the Urinary Stone Disease Research Network (USDRN). The rationale and design of the PUSH trial have been published previously.[17] The protocol was approved by the National Institute of Diabetes and Digestive and Kidney Diseases (NIDDK) and institutional review boards of the Scientific Data Research Center and participating institutions. Patients and members of the public reviewed and provided input on the protocol. Protocol amendments and rationale are detailed in the appendix (p 17). A data safety monitoring board convened by the NIDDK met regularly to assess trial progress and participant safety. Ethics approval was provided by the Duke University Health System Institutional Review Board (Pro0008327).

Participants

English-speaking patients aged 12 years and older with a recent symptomatic urinary stone event were screened. Patients receiving care at USDRN clinical centres and at outside institutions were eligible. Screening, recruitment, and enrolment occurred primarily in connection with clinical care for urinary stone disease. During and after the COVID-19 pandemic, participants could also be screened, recruited, and enrolled remotely. Recent symptomatic stone events were defined as spontaneous stone passage or receiving a procedural intervention to remove stones within the previous 3 years, or a symptomatic stone event within 5 years if new stone(s) were detected on subsequent diagnostic imaging such as ultrasound or CT. Eligible participants were required to have a baseline 24 h urine volume of less than 2·0 L per day for adults and adolescents aged 12–17 years weighing at least 75 kg, and less than 25 mL per kg bodyweight per day for adolescents weighing less than 75 kg. Eligible participants were required to have access to a mobile device that could sync with a Bluetooth-enabled smart water bottle (appendix p 14).[17] Full eligibility criteria are available in the protocol, with the intention of enrolling participants with idiopathic stone disease (ie, not monogenic or from a surgical condition such as bariatric surgery, appendix p 94). Key exclusion criteria included monogenic stone disease, history of hyponatremia, recurrent urinary tract infections, kidney transplantation, anatomic urologic abnormality, or gastrointestinal condition associated with excessive fluid losses. Adult participants provided written informed consent and adolescents provided verbal assent along with consent by legally authorised representatives.

Randomisation and masking

Participants were randomly assigned in a 1:1 ratio to an adaptive multicomponent behavioural intervention focused on increasing fluid consumption (appendix p 5) or a control group with a follow-up of 2 years. Randomisation was stratified by age group (adult or adolescent), stone history (first-time or recurrent stone former), and study site. Randomisation assignment was computer-generated remotely at the Scientific Data Research Center (Durham, NC, USA). Every participant received a Bluetooth-enabled smart water bottle that measured and recorded fluid intake (appendix pp 14–15). Investigators, treating physicians, outcome assessors, and adjudicators were masked to group assignment. Blinding of study participants, coordinators, or health coaches was not possible.

Procedures

At enrolment, participants aged 18 years and older underwent a low-dose, non-contrast CT scan and participants younger than 18 years had a renal ultrasound; all completed a 24 h urine collection. The intervention consisted of a fluid prescription (set measurable behaviour goal BCT BCIO:007300), financial incentives to adhere to fluid prescription, health coaching to overcome barriers to consuming more fluids, and patient-selected approaches such as text messaging to maintain increased fluid intake. These components were selected based on specific behavioural change techniques.[11] The fluid prescription was the additional daily fluid intake in excess of baseline intake required to achieve urine volume of more than 2·5 L per day, as recommended by American Urological Association guidelines,[4] and was calculated with the use of the baseline 24 h urine volume (appendix p 5). The fluid prescription (BCT: goal setting[11] BCIO:007300) was to be consumed from the smart water bottle. Participants in the intervention group were eligible for a daily, loss-framed financial incentive of US$1·50 when they consumed the individualised fluid intake prescription from the smart water bottle (contingency management, BCT: material reward[11]). This financial incentive was

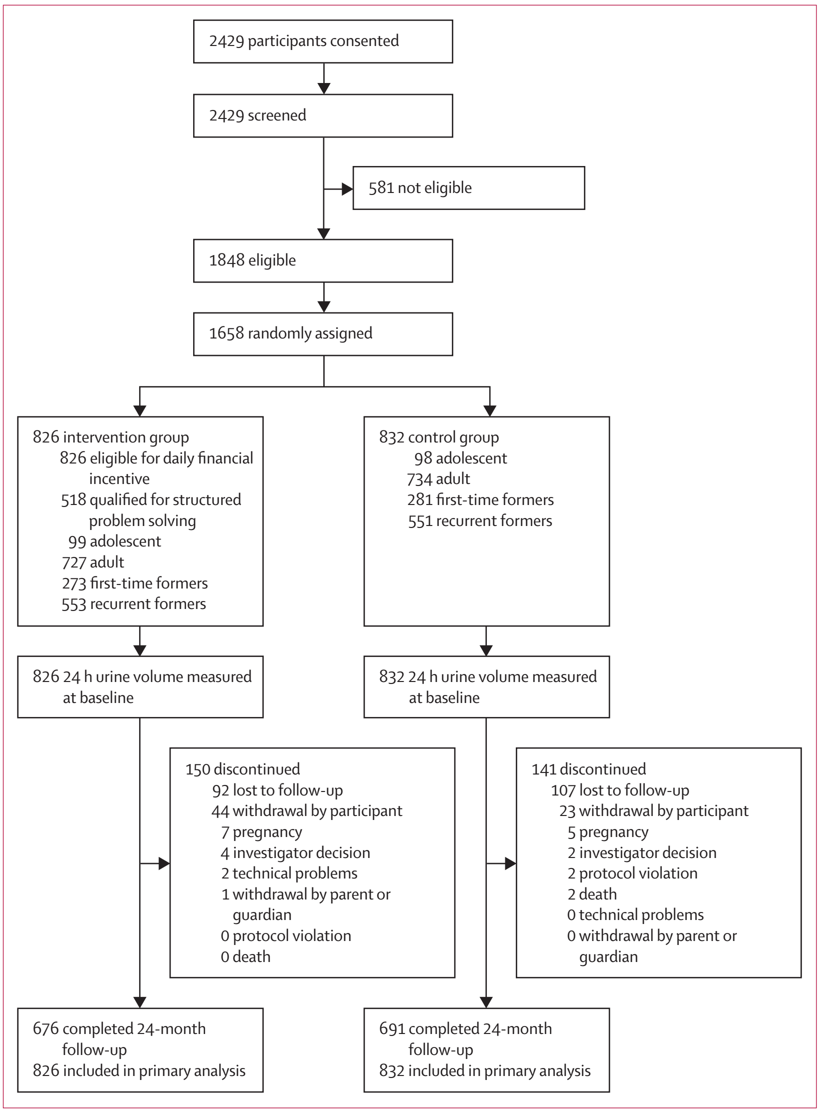


Figure 1: **CONSORT diagram**
All randomly assigned patients included in the primary analysis.

available daily for the first 6 months. In months 6–18, the frequency of the days eligible for financial incentives was tapered from 80% to 15%, and no financial incentives were available in months 19–24 (BCT: provide reduced frequency of aversive consequence for behaviour BCT BCIO:007273).[11,17] Participants randomly assigned to the intervention group were required to synchronise their water bottle and mobile device daily to be eligible for financial incentives. Participants in the intervention group were also offered structured problem solving (health coaching intervention) if the daily fluid prescription was not met for at least two 2-week periods in the first 6 months. Structured problem solving was designed to facilitate identifying barriers to increasing fluid intake and developing solutions to overcome them, prioritising those most practicable (BCT: review behaviour goal [BCIO:007011, action planning BCIO: 007010]).[11,17] A fidelity assessment, conducted by a trained reviewer on a sample of both initial and follow-up coaching interactions, ensured consistency of coaching across institutions.[18] The fidelity assessment showed that, on average, more than 90% of the required elements were covered in each initial and follow-up coaching interaction. Participants received low-touch interventions of their choosing among social incentives (eg, support partner) or low-cost interventions (text message reminders) to help to sustain new habits of fluid consumption during months 19–24 (BCT: social support BCIO:007028, prompts or cues BCIO:007081[11]).

Participants in the control group were provided with guideline-concordant recommendations to increase fluid consumption to achieve urinary output of at least 2·5 L daily in addition to usual stone prevention care.[4,5] They were provided with the smart water bottle, but its use was not required (appendix p 5). All participants in each group continued to receive guideline-concordant care with their kidney stone clinician.

Outcomes

The primary outcome was symptomatic stone recurrence, defined as spontaneous stone passage with symptoms or a procedural intervention for a symptomatic or asymptomatic stone. Participants received questionnaires electronically to report stone events every 3 months for the duration of the study. Reported events were reviewed and classified as confirmed clinical events, participant-reported, or non-events by an adjudication committee comprising expert clinicians who were masked to study group allocation.[19] Confirmed clinical stone events required typical stone symptoms (eg, flank pain) in addition to objective documentation of stone passage or surgical intervention (appendix p 173). Events with typical symptoms and self-described passage of a stone without objective documentation of passage were classified as participant-reported; a detailed description of the adjudication process has been separately published.[19] Confirmed and participant-reported events met the primary outcome. All other reported incidents were considered non-events.

Secondary outcomes included change in 24 h urine volume (a mechanistic outcome); urinary symptoms; radiographic outcomes consisting of new stone formation and growth of an existing stone by at least 2 mm in any dimension; and a composite outcome of symptomatic stone recurrence, new stone formation, or growth of existing stone(s). 24 h urine collections to measure urine volume were done at 6, 12, 18, and 24 months. Participants were considered adherent if they achieved at least two 24 h urine volumes of at least 2·5 L per day for adults or at least 30 mL/kg bodyweight per day for adolescents weighing less than 75 kg at 6, 12, 18, and 24 months. 24 h urine collections were considered adequate if creatinine excretion rate (mg/kg per day) was within two standard deviations of the mean for the cohort. Urinary symptoms were recorded at baseline and every 6 months with the use of the validated Comprehensive Assessment of Self-reported Urinary Symptoms.[20] Radiographical outcomes were assessed with the use of validated software that automated analyses of CT scans.[21] To ensure accuracy and reliability for this trial, the performance of the analysis software was revalidated by study investigators with the use of images from trial participants (appendix p 194).[22] Safety monitoring included report of hyponatremia requiring hospitalisation, which was evaluated systematically by review of all hospitalisations during the study.

Statistical analysis

Statistical analysis was done by personnel masked to treatment assignments. A sample size of 1642 participants (821 per group) provided at least 80% power to detect 30% relative risk reduction in stone recurrence in the intervention group with the use of a log-rank test for time to first symptomatic stone event, with a two-sided type I error rate of 0·05. A 15% event rate was assumed for the control group,[1] and a 20% attrition was assumed over the 24-month follow-up period.

The primary outcome was evaluated according to the intention-to-treat principle and was analysed as time-to-event by treatment group with a log-rank test. Cumulative event rates were calculated for each group as a function of time from randomisation (Kaplan–Meier method). Data collected on subsequent recurrent events of symptomatic stones were analysed by treatment group with the use of the Andersen–Gill model with robust standard errors to account for heterogeneity and correlation between recurrent stone events within a participant.

24 h urine volume measurements at baseline and 6, 12, 18, and 24 months were analysed with a repeated-measures mixed-effects model, with study site treated as a random effect, and treatment and visit as fixed effects. Variance components were used as the variance–covariance structure. The model also included a treatment-by-visit interaction term, and denominator degrees of freedom for the *F*-statistic were approximated with the use of the Kenward-Roger method. Imaging and composite outcomes were analysed as binary outcomes at 24 months with logistic regression models. No interim analyses of the primary and secondary outcomes were planned or performed. No missing outcomes were imputed. For the primary outcome, we used a time-to-first-event approach, with participants who did not have any stone events during the 24-month follow-up censored at the end of their observation

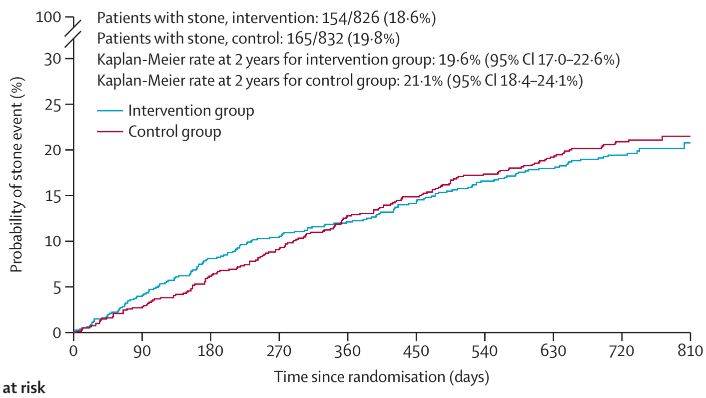
	Intervention group (n=826)	Control group (n=832)	Total (n=1658)
Age, median (IQR), years	45 (30–60)	44 (27–58)	44 (29–59)
Age group, median (IQR), years			
Adult	49 (35–61)	48 (34–59)	48 (36–60)
Adolescent	15 (14–16)	16 (14–16)	15 (14–16)
Female	469 (56·8%)	477 (57·3%)	946 (57·1%)
Male	357 (43·2%)	355 (42·7%)	712 (42·9%)
Race			
White	732 (88·6%)	719 (86·4%)	1451 (87·5%)
Black or African American	55 (6·7%)	58 (7·0%)	113 (6·8%)
Native American	0 (0·0%)	3 (0·4%)	3 (0·2%)
Asian	22 (2·7%)	25 (3·0%)	47 (2·8%)
Native Hawaiian or other Pacific Islander	2 (0·2%)	2 (0·2%)	4 (0·2%)
Other or unknown*	8 (1·0%)	11 (1·2%)	19 (1·1%)
Multiracial	7 (0·8%)	14 (1·7%)	21 (1·3%)
Ethnicity			
Not Hispanic or Latino	761 (92·1%)	757 (91·0%)	1518 (91·6%)
Hispanic or Latino	46 (5·6%)	56 (6·7%)	102 (6·2%)
Not reported	13 (1·6%)	10 (1·2%)	23 (1·4%)
Unknown	6 (0·7%)	9 (1·1%)	15 (0·9%)
Household income			
<\$90 000	323 (39·1%)	305 (36·7%)	628 (37·9%)
≥\$90 000	365 (44·2%)	371 (44·6%)	736 (44·4%)
Other*	138 (16·7%)	156 (18·8%)	294 (17·7%)
Thiazide diuretic at baseline	61 (7·4%)	60 (7·2%)	121 (7·3%)
Potassium citrate at baseline	92 (11·1%)	96 (11·5%)	188 (11·3%)
Median baseline 24 h urine total volume, L, median (IQR), number of adolescents	0·85 (0·66–1·14), 99	0·85 (0·65–1·05), 98	0·85 (0.66–1.10), 197
Median baseline 24 h urine total volume, L, (IQR), number of adults	1·29 (0·98–1·61), 727	1·30 (1·01–1·56), 734	1·30 (1·00–1·58), 1461
Baseline 24 h urine calcium, mg per total volume, median (IQR), number of adults	190 (129–261), 552	190 (127–256), 544	190 (128–259), 1096
Baseline 24 h urine citrate, mg per total volume, median (IQR), number of adults	553 (383–757), 538	534 (376–729), 532	544 (378–746), 1070
Baseline 24 h urine pH, mean (IQR), number of adults	6·00 (5·65–6·41), 671	6·04 (5·66–6·41), 685	6·02 (5·65–6·41), 1356
Baseline 24 h urine osmolality, mOsm/kg, median (IQR), number of adults	662 (471–847), 432	653 (463–815), 469	657 (469–834), 901
Baseline 24 h urine sodium mEq/TV, median (IQR), number of adults	140 (103–183), 549	134 (99–177), 540	137 (101–179),1089
Baseline 24 h urine potassium, mEq/TV, median (IQR), number of adults	50 (36–66), 538	48 (36–62), 525	49 (36–64), 1063

Data are number (%), unless otherwise indicated. mEq/TV=milliEquivalents per total volume. mOsm=milliosmole osmotic concentration. TV=total volume. \$=US\$.*Includes "don't know" and "prefer not to answer" responses.

Table: **Demographic and clinical characteristics of study participants**

period. For the secondary outcome of 24 h urine volume over time (repeated measures), a mixed-effects model was used, and no imputation was required.

Prespecified landmark analyses were done to exclude stone events that occurred within 30, 60, and 90 days of randomisation. These analyses were performed with the use of the same methods as the primary outcome. Planned subgroup analyses for the primary outcome were done with the use of Cox proportional hazards models by including interaction terms for sex, age, clinical centre, provider, and adherence. Participants were considered adherent if they achieved at least two 24 h urine volumes of at least 2·5 L per day for adults or at least 30 mL/kg bodyweight per day for adolescents weighing less than 75 kg at 6, 12, 18, and 24 months. A prespecified sensitivity analysis was done for surgical removal of asymptomatic stones requiring at least one of the following criteria: stone size of at least 4 mm in any dimension, mobile stone, associated haematuria, or recurrent urinary tract infection with the use of the same



Number at risk										
Intervention group	826	770	725	693	663	631	606	588	454	112
Control group	832	791	752	717	674	651	628	608	472	116

Figure 2: **Symptomatic recurrence of urinary stones**

	Number of patients (n)	n/N (%) per 100 patient-years: Intervention group	n/N (%) per 100 patient-years: Control group	$p_{interaction}$
All patients	**1658**	**154 (18·6%) 11·22**	**165 (19·8%) 11·65**	
Stone history				0·34
First time	554	30/273 (11·0%) 6·23	41/281 (14·6%) 7·95	
Recurrent	1104	124/553 (22·4%) 13·91	124/551 (22·5%) 13·77	
Age group				0·33
Adult	1461	139/727 (19·1%) 11·59	144/734 (19·6%) 11·57	
Adolescent	197	15/99 (15·2%) 8·63	21/98 (21·4%) 12·26	
Sex				0·92
Male	712	71/357 (19·9%) 11·91	75/355 (21·1%) 12·59	
Female	946	83/469 (17·7%) 10·69	90/477 (18·9%) 10·97	
Race				0·61
Black or African American	113	8/55 (14·5%) 8·75	11/58 (19·0%) 11·07	
Other	1523	145/764 (19·0%) 11·44	149/759 (19·6%) 11·54	
Stone care provider specialty				0·26
Urologists	1164	107/574 (18·6%) 11·05	125/590 (21·2%) 12·44	
Others	494	47/252 (18·7%) 11·61	40/242 (16·5%) 9·72	
Clinical Centre				0·35
UPenn/CHOP	334	29/168 (17·3%) 9·59	37/166 (22·3%) 12·53	
UTSW	309	24/154 (15·6%) 8·50	29/155 (18·7%) 10·39	
UWash	309	36/154 (23·4%) 15·75	33/155 (21·3%) 13·29	
WUSTL	375	55/186 (29·6%) 19·88	56/189 (29·6%) 18·89	
CCF	198	9/98 (9·2%) 5·40	9/100 (9·0%) 5·04	
Mayo	133	1/66 (1·5%) 0·86	1/67 (1·5%) 0·85	
Adherence based on 2·5 L criteria				0·26
Yes	1124	113/567 (19·9%) 11·19	125/557 (22·4%) 12·62	
No	534	41/259 (15·8%) 11·30	40/275 (14·5%) 9·41	

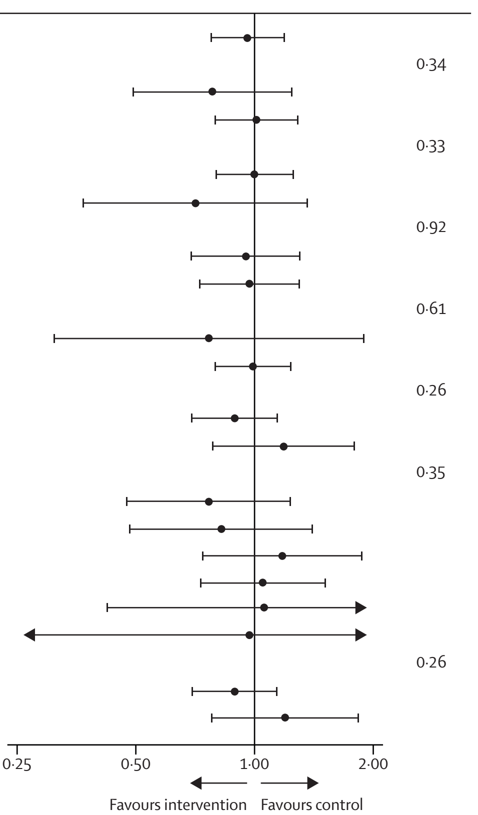


Figure 3: **Subgroup analyses of symptomatic recurrence of urinary stones**
UWash=University of Washington. WUSTL=Washington University in St.Louis. CCF=Cleveland Clinic. Mayo=Mayo Clinic. UPenn/CHOP=University of Pennsylvania/Children's Hospital of Philadelphia. UTSW=University of Texas Southwestern. *Adherence defined as at least 2·5 L urine output per 24 h on at least two 24 h collections for adult participants. Additionally, the 24 h urine collection group is required to have creatinine mg/kg per day within 2 SD of mean Cr mg/kg per day for the cohort.

methods as the primary outcome. As a sensitivity analysis, a win ratio analysis was done with the use of the Finkelstein-Schoenfeld method with the hierarchical order of: (1) symptomatic stone event, (2) new stone formation, and (3) stone growth. A prespecified economic analysis will be published separately.

All statistical comparisons were done with the use of two-sided significance tests with α=0·05 and SAS software (version 9.4). All primary and secondary outcome analyses accounted for the randomisation strata (ie, age group, stone history, and study site). Full analytical plan details are provided in the statistical analysis plan (appendix p 142). This study is registered with ClinicalTrials.gov, NCT03244189.

Role of the funding source

This is a cooperative agreement; that means there is substantial federal scientific or programmatic involvement in the research activities. The NIDDK Project Scientist (ZK) was involved in the design and development of the clinical protocol, preparation of questionnaires and other data recording forms, coordination of research, statistical evaluations and analyses of data, and the publication of results. The programme was overseen by an independent NIDDK Program Official (CM).

Results

2429 patients underwent screening between Oct 23, 2017, and Feb 17, 2022. Of these patients, 1658 were randomly assigned to the intervention group (n=826) or control group (n=832; figure 1). Median age was 44 years (IQR 29–59), and 946 (57%) participants were female (table). 441 (30%) of 1461 of adults and 114 (58%) of 197 adolescents had first-time stone formation. In the intervention group, first-time stone formers included 217 (29·8%) of 727 adults and 57 (57·6%) of 99 adolescents. In the control group, first-time stone formers consisted of 224 ([30·5%] of 734) adults and 57 ([58·2%] of 98) adolescents. Baseline median 24 h urine volume was 1·30 (IQR 1·00–1·58) L per day in adults and 0·85 (0·66–1·10) L per day in adolescents (table).

At a median follow-up of 738 (IQR 711–778) days, 154 (19%) participants had a symptomatic stone event in the intervention group compared with 165 (20%) in the control group (hazard ratio 0·96, 95% CI 0·77–1·20). There was no difference in the cumulative risk of symptomatic stone recurrence between the groups (figure 2). Subgroup analyses showed no heterogeneity of treatment effect by sex, age, clinical centre, or provider (figure 3). In addition, adherence to the intervention, defined as meeting at least two 24 h urine collections at or more than the target value (2·5 L per day for adults), did not result in a difference in the adherence subgroups (figure 3). In sensitivity analyses, there was no difference between groups when excluding individuals who underwent surgery for an asymptomatic stone or stone events that occurred within 30, 60, and 90 days of

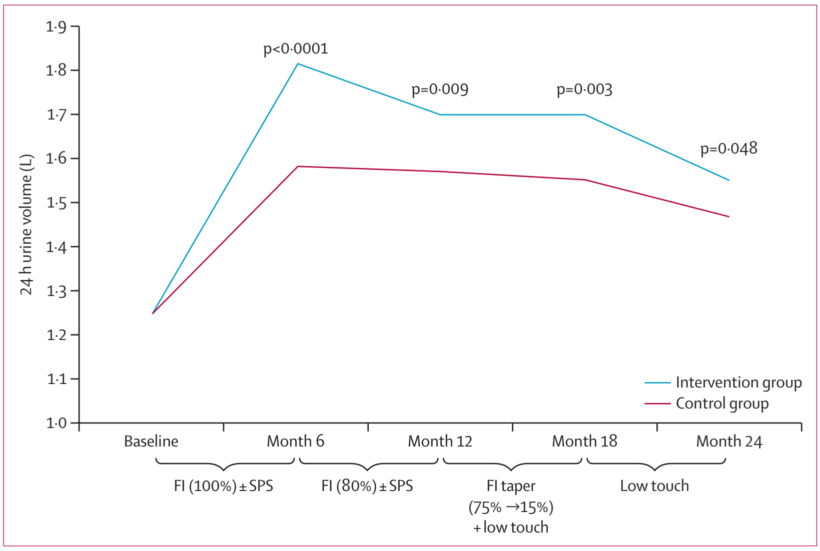


Figure 4: **24 h urine volume by timepoint and treatment group (mixed-effects model)**
FI (%)=loss-framed financial incentive. % indicates percentage of days loss-framed incentive was available to participants. FI Taper=financial incentive taper. Availability tapered from 75% of days (month 13) to 15% of days (month 18). Participants blinded to which day(s) financial incentive was available. No financial incentive available in months 19–24. SPS=structured problem solving. Low touch=low touch interventions to promote adherence (eg, support partner, reminder via text communication).

randomisation (appendix p 9). A hierarchical order (win ratio) analysis did not show a difference between treatment groups (appendix p 10).

24 h urine volume increased from baseline in both groups and was higher in the intervention group at months 6, 12, 18, and 24 compared with the control group (figure 4, appendix p 11). Urinary storage symptoms of frequency, urgency, and nocturia were greater in the intervention group than the control at months 6 (p=0·050) and 12 (p=0·014) but not at other timepoints (appendix p 13). There was no difference in stone growth of at least 2 mm (odds ratio 1·44, 95% CI 0·91–2·29) or new stones (0·99, 0·72–1·37) between groups from baseline to end-of-study imaging (appendix p 12). The composite outcome of symptomatic stone recurrence, new stone formation, or stone growth of at least 2 mm was also not statistically different between groups (appendix p 12).

No participants had hyponatremia requiring hospitalisation. In the intervention group, 12 (1%) participants had asymptomatic hyponatremia versus two (<1%) participants in the control group (p=0·018). There were no study-related serious adverse events in either group.

Discussion

To the best of our knowledge, the PUSH trial is the largest randomised controlled trial designed to improve adherence to fluid intake for secondary prevention of urinary stones to date. There are several key findings. Participants in the behavioural intervention group achieved a greater urine volume compared with participants in the control group. However, this increase in urine volume did not result in a decrease in recurrent symptomatic stone events over 2 years of follow-up. Second, urinary symptoms were greater in the intervention group at 6 and 12 months. Finally, there was no difference between groups in the radiographical outcomes of new stone formation or stone growth at end of study.

Increasing fluid intake to increase urine volume has long been a cornerstone of secondary prevention of urinary stone disease, with the goal of diluting the concentration of stone-forming salts in urine. Both the American Urological Association[4] and the European Association of Urology guidelines[5] recommend fluid intake sufficient to produce a urine volume of at least 2·5 L per day as a prevention strategy for patients with urinary stone disease. These recommendations are partially based on a single randomised controlled trial of 199 participants, which showed that greater fluid intake resulting in 2·5 L of urine per day on average led to fewer recurrent stones and longer time to stone events compared with a group that had an average of only 1 L of urine per day.[7] Largely based on the reduction in recurrent stone events reported in that trial, increased fluid intake is now considered standard for the secondary prevention of urinary stone disease.[4,5] However, in real-world care, it is challenging for patients to increase their urine volume to meet this goal, despite the efforts of various providers in multiple clinical settings.[8]

In the PUSH trial, adult participants in the intervention group increased urine volume (*vs* baseline) by 600 mL per 24 h at 6 months, compared with only 360 mL per 24 h in the control group, but this difference diminished over time. An important aspect of the PUSH trial was that participants in the control group continued to receive ongoing care from their existing stone clinician, which included increasing fluid intake and dietary recommendations per their usual practice, compared with the Borghi and colleagues' trial,[7] in which participants in the control group received no additional fluids. Accordingly, the increase in urine volume in the control group in our study and the resulting smaller difference in urine volume between the two groups (figure 4) might explain the similarity between groups for stone-related outcomes.

There are myriad perceived barriers to fluid intake such as not liking the taste of water, not experiencing thirst, failure of habit formation for fluid intake, and feeling bloated with consumption. Additionally, many people do not understand the association between fluid intake and stone formation.[9] In this study, we attempted to address barriers to fluid intake through health coaching and structured problem solving, theorising that patient education and providing individualised solutions would optimise success of the intervention. However, the difference in urine volume between groups remained modest. This outcome might, in part, reflect a unique negative effect of the intervention for this trial in that urinary symptoms are often exacerbated by increased fluid intake. Participants assigned to the intervention group reported greater urinary symptoms at months 6 and 12—timepoints at which the difference in urine volume between the two groups was greater (appendix p 13). As fluid intake increases, urine output is expected to rise, which in some patients might result in expected, but potentially bothersome, urinary symptoms such as frequency or nocturia. These parallel increases in urine volume and urinary symptoms might reflect the challenges that affect some people who form stones when trying to adhere to an increased fluid intake goal.

Findings from several randomised controlled trials have shown that interventions can improve healthy behaviours such as smoking cessation, weight loss, and physical activity.[12,23,24] Broad adoption of these approaches has been hindered by two key limitations: durability of the health behaviour change and the use of surrogate rather than clinically meaningful outcomes. We designed the PUSH trial to address these limitations.[17] First, we used a clinically meaningful primary outcome (symptomatic stone recurrence) rather than the mechanistic surrogate outcome of increased urine volume.[25] Second, we structured the intervention to enhance durability of the effect through habit formation with the use of behavioural approaches known to improve adherence, including an initial loss-framed incentive, tapering of incentives with intermittent reinforcement,[11,26] and additional components that operate on structural barriers not addressed by financial incentives. Adaptable patient-selected strategies allowed for individualisation of plans to maximise success of the intervention. The positive effect on a surrogate, but less patient-relevant outcome (24 h urine volume) highlights the importance of following interventions to meaningful clinical endpoints in adherence trials and should inform the design of future health behaviour trials.

Our study has limitations. Participants were recruited from academic, tertiary care centres that might treat participants with more severe disease. We did not assess or control for additional fluid intake not measured by the smart water bottle or other dietary factors, although randomisation should balance these unmeasured confounders. The follow-up period was only 2 years; however, the PUSH trial exceeded the estimated event rate, and the Kaplan–Meier curves were nearly parallel, suggesting that a longer follow-up period for PUSH is unlikely to change the results. Insight into what components of the intervention were effective (or not) would have required a larger study design (eg, factorial), which was not feasible. These factors could be assessed in an exploratory secondary analysis. Females made up 57% of the study population. Although the prevalence of urinary stone disease among females in the USA is closer to 44%, the gender gap is narrowing worldwide.[27] The predominance of female participants might reflect gender differences in willingness to participate in clinical research, as females are more likely than males to enrol in behavioural intervention trials[28] (median 56·7%, 95% CI 40·7–76·0% [n=1346 trials]). Utilisation of adjunctive stone prevention medications (ie, thiazides and potassium citrate) was similar between groups at baseline (table). We did not assess participant acceptability with the intervention, although the dropout rate was lower than projected. Finally, there is the risk of co-intervention through treatment in a specialty stone clinic, known as the stone clinic effect.[29] This effect, along with frequent study contact with the control group participants, could have biased results toward the null hypothesis. These limitations notwithstanding, several features of the trial strengthen the validity of outcomes. The study population was the largest to date for a stone prevention trial, with adequate power to detect meaningful differences, and the number of primary outcome events exceeded projections. Additionally, we exceeded the target enrolment, with an attrition rate lower than projected. Multiple complementary methods were used to ascertain outcomes to reduce under-reporting of events.

Although there was no difference in stone recurrence between the groups, our trial has important implications for behavioural science and for stone prevention efforts. This intervention increased urine volume but did not decrease the long-term clinical goal of stone recurrence. The results of the PUSH study do not undermine the importance of increasing fluid intake for stone prevention, as this remains a low-cost, low-risk intervention with likely benefits based on previous literature. Rather, our study suggests that although contingency management intervention with health coaching for non-adherent participants was insufficient to yield significant clinical benefit, relative to participants in the control group, future investigations could apply optimisation study designs (ie, factorial designs, or sequential, multiple assignment, randomised trial designs) within a multiphase optimisation strategy framework to determine what strategies yield the best results for most people. Additional qualitative research might yield key insights into patient barriers to fluid intake that would inform future intervention design. A more personalised medicine approach could be attempted to address individual differences in motivations to consume water and thereby prevent recurrent stones.

Urinary Stone Disease Research Network

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Contributors

ACD, JDH, GET, and CDS: study design, investigation, data collection, analysis plan, writing of the original draft, review and editing, and funding acquisition. NMM, HHL, PPR, and HW: study design, investigation, data collection, analysis plan, review and editing, and funding acquisition. SS and JCL: investigation, data collection, analysis plan, review and editing, and funding acquisition. HY: study design, data curation, formal analysis, writing of the original draft, and review and editing. HRAl-K: study design, investigation, data collection, analysis plan, formal analysis, review and editing, and funding acquisition. ZK: study design, analysis plan, review and editing, and supervision. CDS and HRAl-K had full access to all the blinded data in the study. HRAl-K and HY had direct access to and verified the underlying data and take responsibility for the integrity of the data analysis. All authors contributed to the drafting and review of the manuscript and agreed to submit it for publication. All authors had permission to access the raw data in this study.

Declaration of interests

JCL declares grants or contracts from Alnylam, Dicerna/Novo Nordisc, Arbor, Oxalosis, and Hyperoxaluria Foundation; consulting fees (paid to

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Data sharing

After publication, de-identified data from the trial database will be transferred to the NIDDK Central Data & Biorepository (https://repository.niddk.nih.gov/home). Proposals for data access can be submitted via email (USDRN@duke.edu) or directly to the NIDDK Central Data & Biorepository. Data will be made available to those with an approved proposal and executed data access agreement.

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